

Physics Informed Priors for MLIPs with UQ

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1 Meta / abstract points

1. To propagate uncertainties up scales we need to capture MLIP errors at the level of MLIP parameters
2. However, the resultant model ensemble can be highly unphysical.
3. Here, we implement physics informed priors for linear MLIPs through inequality constraints
4. We provide an open source implementation within the POPS UQ scheme
5. For simple models, the constraints notably improve phonon stability
6. The final scheme provides robust max/min bounds while retaining physically meaningful model parameters.

2 Main points

1. To communicate that addressing holes in the potential energy surface is traditionally addressed by evaluating more DFT (via hyper-active learning, fine-tuning, etc.) and can be done less expensively
2. DFT stresses are not well converged so elastic properties are often incorrectly predicted by DFT
3. Non-repulsive core interactions for potentials are often observed, using our priors allows us to access high pressure conditions
4. Priors can be formulated based on experimental data
5. Inclusion of UQ via POPS and analysis of how this affects coverage of training data/test data

3 Method

3.1 Linear MLIPs and the constrained fit

Points to make.

- Linearity in the parameters is the structural property the whole scheme rests on — it is what makes every physical observable below an *exact* linear functional of c , and hence expressible as a QP row.
- Contrast with committee schemes for non-linear MLIPs, where physical constraints can only be imposed by penalty or by retraining.

For an atomic configuration R the ACE energy is linear in the parameter vector $c \in \mathbb{R}^d$,

$$E(R; c) = \sum_{k=1}^d c_k B_k(R) = B(R) \cdot c, \quad (1)$$

with B_k the ACE basis functions. Because differentiation is linear, every derivative of E inherits the same affine dependence on c ; in particular the force-constant matrix

$$\Phi_{a\alpha, b\beta}(c) = \frac{\partial^2 E}{\partial r_{a\alpha} \partial r_{b\beta}} = \sum_{k=1}^d c_k \Phi_{a\alpha, b\beta}^{(k)}, \quad \Phi^{(k)} = \frac{\partial^2 B_k}{\partial r \partial r}, \quad (2)$$

splits into d geometry-only tensors. We refer to $\{\Phi^{(k)}\}$ as the *undotted* Hessian: it depends on the geometry but not on c , so it is built once per geometry and reused for every ensemble member. This is what reduces a per-member phonon calculation from a full Hessian build to a matrix–vector product and a small eigendecomposition, cheap enough to serve as a rejection predicate. `scripts/bandpath_phonon_uq/lib.jl` (`undotted_hessian`, `undotted_Hbasis`).

The unconstrained fit is ridge regression,

$$\text{minimize} \quad \|Xc - y\|_2^2 + \|\Gamma c\|_2^2, \quad (3)$$

with X the design matrix and y the observations (energies, forces, virials). Linear constraints are imposed by writing this as a quadratic program,

$$\begin{aligned} &\text{minimize} \quad \frac{1}{2} c^T P c + q^T c \\ &\text{subject to} \quad l \leq A c \leq u, \end{aligned} \quad (4)$$

where $P := X^T X + \Gamma^T \Gamma \in \mathbb{S}_+^d$ and $q := -X^T y$. Each row of A with its bounds (l, u) is one physical requirement; an equality is the case $l = u$. We solve Eq. (4) with OSQP.

Preconditioning, and what the prior actually is. The fit is carried out in a preconditioned variable. With W the diagonal observation weights, S the diagonal preconditioner and $\lambda = 1/M$ (M the number of observations),

$$\tilde{X} = WX S^{-1}, \quad \tilde{y} = Wy, \quad \tilde{c} = Sc, \quad (5)$$

and the objective actually solved is $\frac{1}{2}\|\tilde{X}\tilde{c} - \tilde{y}\|_2^2 + \frac{\lambda}{2}\|S\tilde{c}\|_2^2$. Substituting $\tilde{c} = Sc$ shows the effective prior in Eq. (3) is

$$\Gamma = \sqrt{\lambda} S^2, \quad (6)$$

i.e. the *square* of the preconditioner. A constraint row g acting on c is transported as gS^{-1} acting on $\tilde{c}$, and the solution recovered as $c = S^{-1}\tilde{c}$. **[TODO: decide whether Eq. (6) belongs in the main text or a footnote. It is a property of the implementation, but a referee reproducing the prior will otherwise infer $\Gamma = S$ and get a different regulariser.]**

3.2 POPS ensembles

Points to make.

- POPS is *post hoc*: one fit, then closed-form perturbations. No retraining per member, which is what makes constraining every member affordable.
- Each member answers a counterfactual — “what would the fit have been had this observation been reproduced exactly?”
- The min/max envelope over members is a range statistic, not a confidence interval. Say so explicitly to pre-empt the obvious referee question.

Write $C = \tilde{X}^T \tilde{X} + \lambda S^2$, $G = C^{-1} \tilde{X}^T$, leverage $h_i = (\tilde{X}G)_{ii}$ and residual $r = \tilde{y} - \tilde{X}\tilde{c}_{\text{LSQ}}$. The POPS pointwise correction and corresponding ensemble member are

$$\delta c^{(i)} = S^{-1}G_{:,i} \frac{r_i}{h_i}, \quad c^{(i)} = c_{\text{LSQ}} + \delta c^{(i)}. \quad (7)$$

The correction is exactly the perturbation that makes the model interpolate observation i ,

$$\tilde{X}_{i,:}(S\delta c^{(i)}) = \tilde{X}_{i,:}G_{:,i} \frac{r_i}{h_i} = h_i \frac{r_i}{h_i} = r_i, \quad (8)$$

so the member’s prediction at i moves by precisely the residual there. `src/POPS/POPSRegression.jl` (`corrections`); the per-observation construction used here agrees with the module to 8×10^{-8} .

Members are drawn by an axis-aligned hypercube proposal in the eigenbasis of the correction cloud. Collecting corrections as rows of Δ , diagonalising $\Delta^T \Delta = V \Lambda V^T$ and retaining directions with $\Lambda_{jj} > 10^{-8} \max_l \Lambda_{ll}$, the box is the bounding box of the projections $\Pi = \Delta V$,

$$\ell_j = \min_i \Pi_{ij}, \quad u_j = \max_i \Pi_{ij}, \quad (9)$$

with proposals $c = c_0 + V(\ell + (u - \ell) \odot \mathbf{u})$, $\mathbf{u} \sim \mathcal{U}[0, 1]^p$. The box is *unclipped*: the feasible-set geometry is imposed exactly by rejection rather than approximated by shrinking the proposal. `src/POPS/POPSRegression.jl` (`hypercube`, `rejection_sample_hypercube`).

A rank remark. A cloud of N members spans at most $N - 1$ directions, so Eq. (9) confines the ensemble to a subspace of dimension $\leq N - 1$ of the d -dimensional parameter space. For $N \ll d$ the predictive envelope is systematically too narrow — quantified in Sec. 4.9.

3.3 Physical constraints

Points to make.

- Each constraint is a statement about the potential energy surface, not about the data. That is the sense in which they are priors.
- All are exact linear rows in c : no finite differences, no retraining, no penalty parameters to tune.
- They are cheap relative to acquiring more DFT, which is the alternative route to the same physics.

Equation of state. Let $b(a) = B(\text{reference cell at lattice constant } a)$, so that $E(a; c) = b(a) \cdot c$ is again linear. With $b' = db/da|_{a_{\text{eq}}}$ and $b'' = d^2b/da^2|_{a_{\text{eq}}}$,

$$b' \cdot c = 0 \quad (\text{equality}), \quad b'' \cdot c \geq \varepsilon, \quad (10)$$

states that a_{eq} is a stationary point of the energy–volume curve and that it is a minimum. Together these pin every member’s equilibrium lattice constant to a common reference, which is what makes a *fixed* reference geometry legitimate for the phonon rows below.

Born elastic stability. The Voigt elastic operator is linear in c , $C_{ij}(c) = \sum_k c_k H_{ij,k}^{\text{el}}$ with $H_{ij,k}^{\text{el}} = \partial^2 B_k / \partial \varepsilon_i \partial \varepsilon_j$, so the cubic Born criteria enter as three rows,

$$C_{44} > 0, \quad C_{11} - C_{12} > 0, \quad C_{11} + 2C_{12} > 0. \quad (11)$$

`src/Elasticity/get_hessian.jl` (`elastic_hessian_basis`).

Phonon stability. From Eq. (2) the Bloch transform to the primitive-cell dynamical matrix is a fixed linear map, so

$$D(\mathbf{q}; c) = \sum_{k=1}^d c_k D^{(k)}(\mathbf{q}), \quad (12)$$

with

$$D_{i\alpha,j\beta}^{(k)}(\mathbf{q}) = \frac{1}{\sqrt{m_i m_j}} \sum_{l: \text{s2p}(l)=j} \Phi_{\text{p2s}(i)\alpha, l\beta}^{(k)} e^{i\mathbf{q} \cdot \mathbf{R}_l}, \quad (13)$$

where $\mathbf{R}_l$ is taken under the minimum-image convention in supercell fractional coordinates. Frequencies follow from the eigenvalues μ_ν of D ,

$$\omega_\nu(\mathbf{q}; c) = \text{sgn}(\mu_\nu) \sqrt{|\mu_\nu|} f, \quad f = \frac{1}{2\pi \times 10^{12}} \sqrt{\frac{e}{\text{\AA}^2 u}} \approx 15.633 \text{ THz}, \quad (14)$$

the signed square root displaying imaginary (unstable) modes as negative frequencies. `src/Phonons/phonon_bands.jl` (`precompute_force_constants`, `dynamical_matrix_from_fc`); band-path assembly in `scripts/bandpath_phonon_uq/lib.jl` (`bandpath_Dk`).

The scalar stability metric used throughout is the minimum non-acoustic frequency over the sampled high-symmetry path,

$$m(c) = \min_{\mathbf{q}: \|\mathbf{q}\| \geq q_r} \min_{\nu} \omega_\nu(\mathbf{q}; c), \quad (15)$$

excluding a neighbourhood of Γ where the acoustic branches vanish by construction.

Cutting planes. Requiring $\min_{\nu} \mu_\nu \geq \omega_{\text{cut}}^2$ is not linear in c , but it is the infimum of linear constraints. For any fixed unit vector e the Rayleigh quotient is an *exact* linear functional of c ,

$$e^\dagger D(\mathbf{q}; c) e = \sum_k c_k \underbrace{\text{Re}[e^\dagger D^{(k)}(\mathbf{q}) e]}_{g_k(\mathbf{q}, e)} = g(\mathbf{q}, e) \cdot c \geq \omega_{\text{cut}}^2, \quad (16)$$

requiring no finite differences and no displacement amplitude. Since $e^\dagger D e \geq \mu_{\min}(D)$ for every unit e , one such row is necessary but not sufficient: the eigenvector rotates as c moves. The constraint set is therefore accumulated by a cutting-plane loop — re-diagonalise, append rows for whatever remains soft, re-solve — until no soft mode remains. `scripts/bandpath_phonon_uq/lib.jl` (`cut_row`, `soft_modes`).

3.4 Constraining across volumes

Points to make.

- A curvature condition at one geometry says nothing about the stretched lattice a barostat actually visits. This is the central methodological finding of the paper.
- The extra volumes add stability requirements only; they do not move the equilibrium, which stays pinned by Eq. (10).
- Cost is one undotted Hessian build per volume, amortised over the entire ensemble.

Phonon positivity is imposed not at a_{eq} alone but on a grid spanning the thermal-expansion range,

$$a_v = s_v a_{\text{eq}}, \quad s_v \in \{1.00, 1.02, 1.04, 1.06, 1.08, 1.10\}. \quad (17)$$

Each a_v carries its own $\{\Phi^{(k,v)}\}$ and $\{D^{(k,v)}(\mathbf{q})\}$; the cutting-plane loop pools soft modes over all v , and acceptance requires $m_v(c) \geq \omega_{\text{cut}}$ for every v . `scripts/uq/bandpath.committee.undotted`

3.5 Rejection sampling, and a cheaper scheme

Points to make.

- Rejection cannot impose an equality — zero acceptance probability under a continuous proposal — so the equality lives in the QP and its relaxation in the predicate. Readers will otherwise trip on this.
- The cheap variant is the practical contribution: physics enters through the *sampler* rather than through a per-observation QP.

A proposal is accepted if it satisfies the Born rows Eq. (11), phonon positivity at every volume, and the band

$$\frac{|b' \cdot (c - c_0)|}{b'' \cdot c_0} \leq \delta, \quad (18)$$

which to leading order states $|\Delta a_{\text{eq}}| \leq \delta$, since $\Delta a_{\text{eq}} \simeq -(dE/da)/(d^2E/da^2)$. Eq. (18) is the continuous relaxation of the equality in Eq. (10).

This suggests a substantially cheaper construction. Rather than solving a cutting-plane QP carrying $O(10^3)$ phonon rows for *every* observation, we constrain each member only on the equation of state — one small QP — and let the rejection sampler enforce phonon positivity using the precomputed $\{D^{(k,v)}\}$:

1. for each observation i , solve Eq. (4) with the interpolation row $\tilde{X}_{i,:} \tilde{c} = \tilde{y}_i$ and the rows of Eq. (10), and *no* phonon rows;
2. fit the hypercube Eq. (9) to the resulting cloud;
3. accept draws satisfying Eq. (11), Eq. (18), and $m_v \geq \omega_{\text{cut}}$ for all v .

`scripts/uq/aeq_constrained_pops_coverage_A1.12.4.6A.2.jl`.

3.6 Validation observables

Molecular dynamics. NPT at zero pressure (Langevin thermostat, Monte-Carlo barostat), $4 \times 4 \times 4$ FCC supercell, $T \in \{300, 500, 700, 900\}$ K. The lattice constant and linear expansion coefficient are

$$a(T) = \frac{1}{|\mathcal{P}|} \sum_{f \in \mathcal{P}} \frac{V_f^{1/3}}{N_{\text{sup}}}, \quad \alpha = \frac{1}{a_0} \frac{da}{dT}, \quad (19)$$

averaged over production frames $\mathcal{P}$. Because the cell breathes under the barostat, every pair analysis uses each frame’s own box for the minimum-image convention.

Structural integrity. Thermal expansion alone does not establish that the lattice is still the one you started from. We therefore report the mean coordination within a cutoff r_c ,

$$\langle Z \rangle = \frac{1}{N} \sum_i |\{j \neq i : r_{ij} < r_c\}|, \quad \text{FCC iff } \langle Z \rangle \geq 12 - \tau, \quad (20)$$

together with the radial distribution function, time-averaged over production frames with per-frame boxes. α in Eq. (19) is fitted *only* over temperatures satisfying Eq. (20). **[TODO: state in the text that the *mean* coordination is required — the maximum remains at 12 long after the lattice has transformed, which is how a transformed cell can be mistaken for thermal expansion.]**

Grüneisen parameters. Following Togo and Tanaka, the mode Grüneisen parameter is

$$\gamma_{\mathbf{q}j}(V) = -\frac{V}{\omega_{\mathbf{q}j}} \frac{\partial \omega_{\mathbf{q}j}}{\partial V} = -\frac{V}{2\omega_{\mathbf{q}j}^2} \sum_{\alpha\beta, jj'} e_{\mathbf{q}j}^{\alpha j*} \frac{\partial D_{jj'}^{\alpha\beta}(V, \mathbf{q})}{\partial V} e_{\mathbf{q}j}^{\beta j'}, \quad (21)$$

the second form being branch-safe because it contracts $\partial D/\partial V$ against the reference-volume eigenvector rather than differencing sorted frequencies. The macroscopic parameter follows by heat-capacity weighting, $\gamma = \sum_{\mathbf{q}j} \gamma_{\mathbf{q}j} C_{\mathbf{q}j} / C_V$. `scripts/uq/gruneisen_constrained_vs_naive_A1.12.4.6A.2.jl`.

4 Results

All results are for a linear ACE model of aluminium (A1.12.4.6A.2., $d = 91$ parameters, $M = 146\,958$ observations, 6 Å cutoff). **[TODO: give the basis specification properly: body order, polynomial degree, radial basis, and the training set provenance.]**

4.1 Phonon stability at the equilibrium geometry

The naive POPS ensemble is catastrophically unstable: members exhibit imaginary frequencies over much of the Brillouin zone despite each being a legitimate perturbation of a well-fitted model. Imposing Eqs. (10), (11) and (16) removes this entirely (Fig. 1).

Points to make.

- Quote the fraction of members with $m(c) < 0$: 30/30 naive versus 0/30 constrained.
- Emphasise that no additional DFT was used — the improvement comes entirely from the prior.

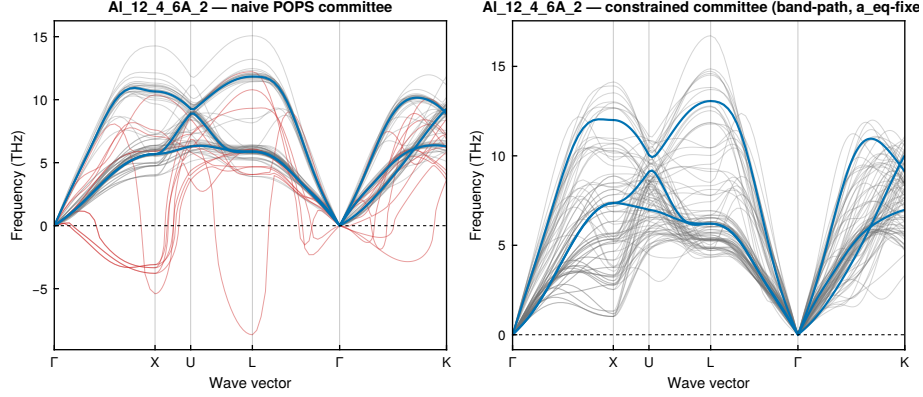


Figure 1: a) Ridge regression solution of Al ACE model with POPS hypercube ensemble. b) Constrained ridge regression solution with constrained POPS hypercube ensemble. The constraints are that the phonon energies along the band-path are greater than 0 eV.

- [TODO: state the supercell convergence explicitly: $N_{\text{cell}} = 3$ is *not* converged for a 6 Å cutoff (box half-width $\approx$ cutoff gives zone-boundary aliasing); all results here use $N_{\text{cell}} = 4$, verified against $N_{\text{cell}} = 5$.]

4.2 Stability across the thermal-expansion range

Constraining at a_{eq} alone is insufficient. The softest member of the single-volume constrained ensemble is stable at a_{eq} ($m = +0.278$ THz) yet leaves the FCC lattice within picoseconds of NPT at 300 K, with $\langle Z \rangle$ falling from 12.00 to 9.5 and $\Delta E \approx -0.18$ eV/atom. Imposing Eq. (17) across six volumes gives 0/30 members unstable at any volume against 30/30 for the naive ensemble.

Table 1: Minimum non-acoustic frequency (THz) over the ensemble at each constrained volume; cf. Fig. 2.

s_v	1.00	1.02	1.04	1.06	1.08	1.10
naive	−8.656	−10.920	−11.471	−10.812	−9.043	−9.077
constrained	+0.252	+0.175	+0.150	+0.166	+0.150	+0.153

Points to make.

- The soft mode is *not* monotonic in volume: $s = 1.04$ binds more strongly than $s = 1.06$. Constraining only the endpoints would miss an interior soft spot — this justifies a grid rather than a two-point bracket.

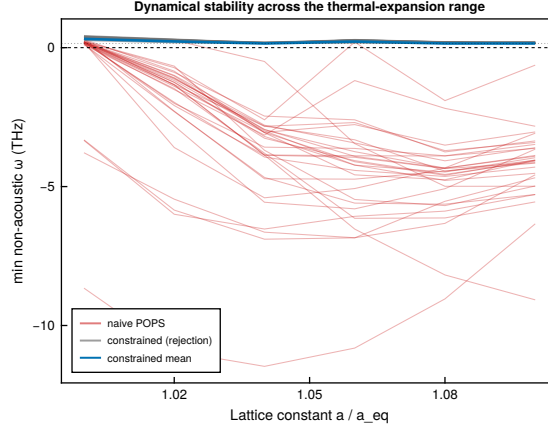


Figure 2: Minimum non-acoustic frequency against lattice constant across the constrained range. Each red curve is one naive POPS member, each grey curve one constrained + rejection-sampled member, and the blue curve the constrained mean. The dotted line is the constraint margin ω_{cut} . Every naive member crosses zero somewhere in the range; no constrained member does.

- The cutting plane drives members onto the margin ω_{cut} , so *which* member is softest is near-degenerate and not reproducible between runs, while the margin itself is (worst-over-volumes agrees to 10^{-4} THz between independent runs). Quote ensemble statistics, not individual member indices.

4.3 Finite-temperature structural stability

Points to make.

- These two figures are the core qualitative result: identical analysis, identical temperature, opposite verdict.
- Panel (c) is the cleanest single discriminator — a spike at 12 versus a smear centred on 9.
- **[TODO: state that the transformed phase is neither molten (MSD sub-diffusive, plateauing near $0.9 \text{ \AA}^2$) nor cavitated (a probe-grid scan finds no voids); it is a homogeneous lower-coordination solid at larger volume.]**

4.4 Thermal expansion

Points to make.

- The single-volume value (18.95) looks respectable but is fitted entirely over a transformed lattice — a cautionary example of why Eq. (20) is needed. Its LOO range spans a factor of two.

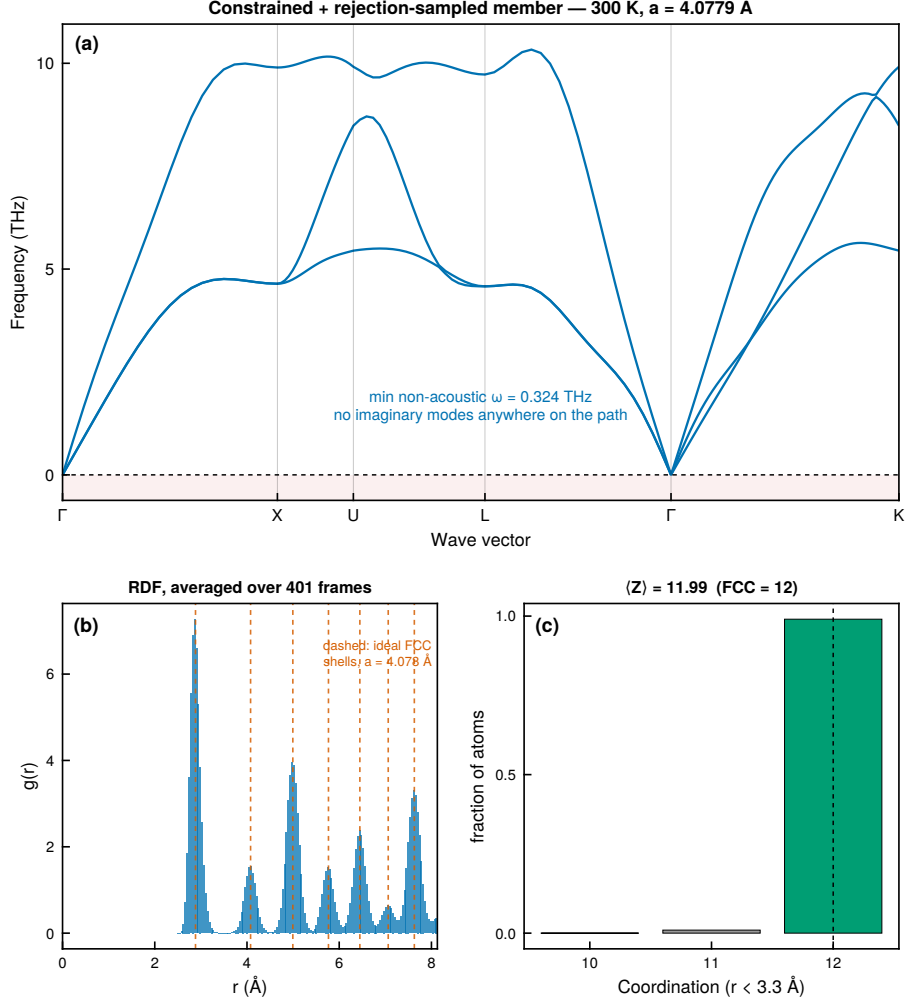


Figure 3: Constrained and rejection-sampled member at 300 K. (a) Phonon bands at the NPT lattice constant $a(300\text{ K})$, entirely positive with $\min \omega = +0.324$ THz. (b) Radial distribution function, time-averaged over 401 production frames, with ideal FCC shells $r_n = a\sqrt{n/2}$ marked. (c) Per-atom coordination distribution: a sharp spike at $Z = 12$.

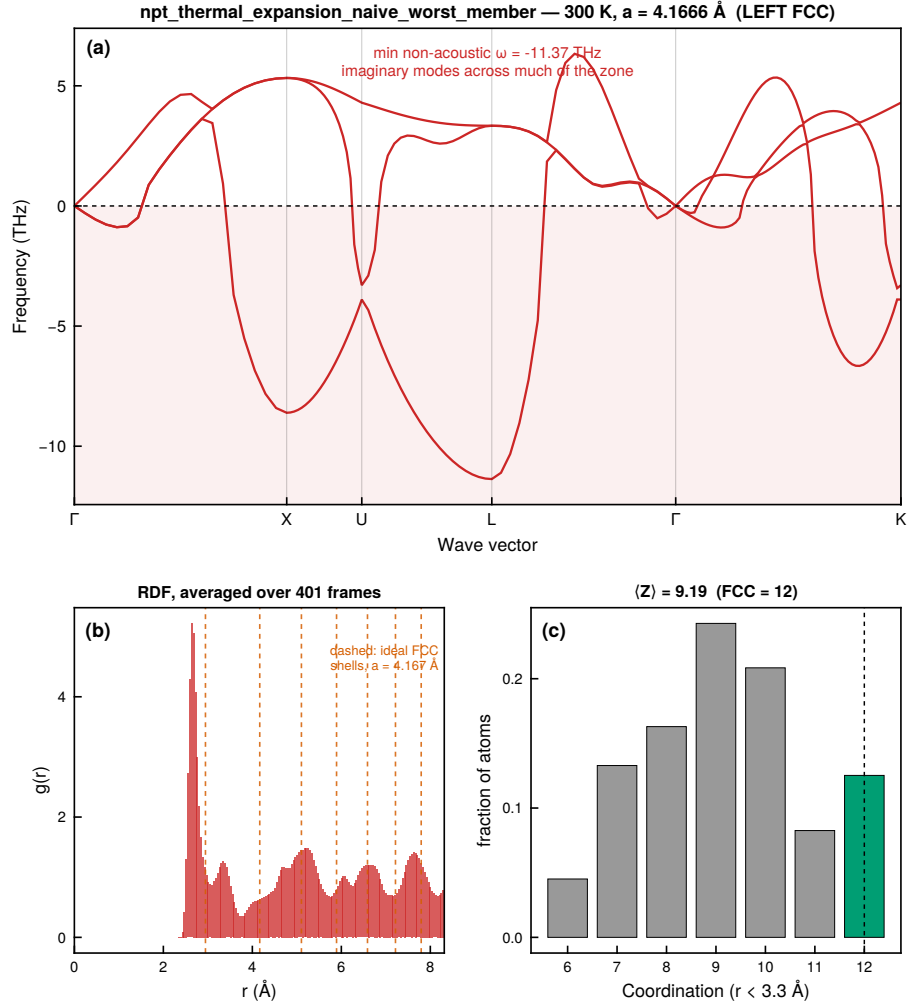


Figure 4: As Fig. 3 but for the worst unconstrained (naive) POPS member at 300 K: imaginary modes across much of the zone ($\min \omega = -11.37$ THz), an RDF whose peaks do not coincide with any FCC shell, and coordination smeared over $Z = 6$ –12 with $\langle Z \rangle = 9.19$.

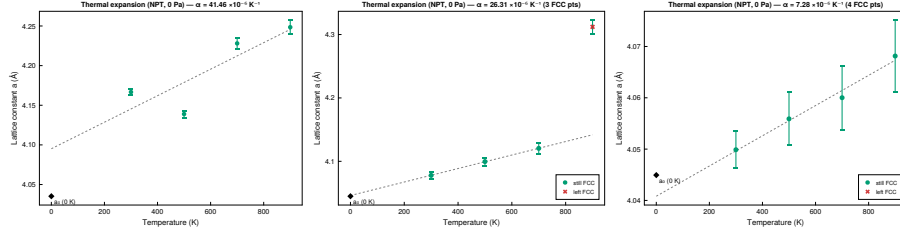


Figure 5: $a(T)$ from NPT at zero pressure. Left: worst naive POPS member. Centre: multi-volume constrained member. Right: cheap EOS-QP + rejection member. Filled circles are temperatures at which the lattice remained FCC by Eq. (20); crosses are temperatures at which it did not, and are excluded from the fit. The diamond marks the 0 K relaxed a_0 . **[TODO: the naive and single-volume panels have a_0 far off the fitted line precisely because every finite- T point is a transformed lattice — worth pointing out in the caption as the visual signature of the failure.]**

Table 2: Linear thermal expansion coefficient and FCC survival. Leave-one-out (LOO) ranges indicate the sensitivity of the fit to individual temperatures. Experiment: $\alpha \approx 23 \times 10^{-6} \text{ K}^{-1}$ at 300 K rising to $\approx 26 \times 10^{-6}$ by 700 K.

member	α (10^{-6}K^{-1})	LOO range	FCC survival
naive, worst	41.46	[34.4, 67.9]	0/4
constrained, single volume	18.95	[14.8, 31.7]	0/4
constrained, multi-volume	26.31	[25.8, 26.8]	3/4
cheap: EOS-QP + rejection	7.28	[6.26, 7.56]	4/4

- The multi-volume member reproduces α to within a few per cent of experiment with a LOO range of $\pm 2\%$ and $R^2 = 0.99988$.
- The cheap variant is the most *stable* (4/4 FCC, including 900 K where the multi-volume member fails) but the least accurate in α : it is over-stiff. Frame this as a stability/anharmonicity trade-off — its phonon margin is larger (+0.255 against +0.150 THz worst-over-volumes), and a stiffer potential both holds FCC better and under-expands.
- The failure at 900 K of the multi-volume member is *not* a harmonic instability: $a(900 \text{ K}) = 1.066 a_{\text{eq}}$ lies inside the constrained grid and its phonons there are positive (+0.208 THz). It is a finite-amplitude, thermally activated transition, which harmonic constraints cannot address by construction. 900 K is also 96% of the melting point.
- **[TODO: decide how hard to push the trade-off framing — it is currently $n = 1$ per arm. The ensembles below give it error bars.]**

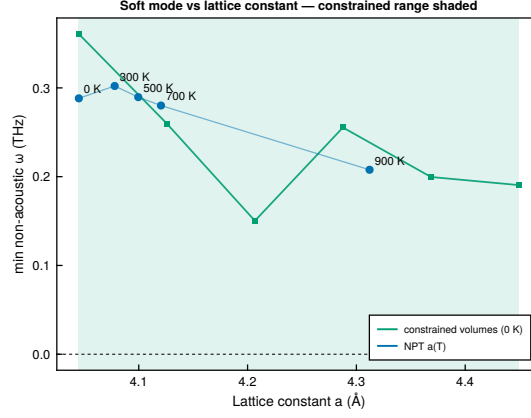


Figure 6: Soft-mode frequency against lattice constant for the multi-volume constrained member. The shaded region is the constrained range Eq. (17); squares are the 0 K constrained volumes and circles the lattice constants visited under *NPT*. The 900 K point lies inside the constrained range with positive phonons, confirming that the loss of FCC there is finite-amplitude rather than a harmonic instability.

4.5 Ensemble thermal expansion with POPS uncertainties

[TODO: Jobs 6000161 (constrained) and 6000267 (naive) were still running at the time of writing; the numbers below are placeholders. Aggregated by scripts/uq/npt_ensemble_aggregate_A1.12.4_6A.2.jl.]

Six trajectories were run per ensemble — the constrained mean (respectively the regularised least-squares model) plus five members — with *identical* MD seeds at each temperature, so that the spread across members is attributable to the parameter vectors rather than to MD noise.

Table 3: Ensemble thermal expansion. **Placeholder — awaiting jobs 6000161 / 6000267.**

ensemble	α mean $\pm$ s.d. (10^{-6}K^{-1})	range	members retaining FCC
constrained + rejection	—	—	—
naive POPS	—	—	—

Points to make.

- This is the headline UQ claim: constrained + rejection yields a bounded, physical interval for α , whereas naive POPS does not — for members that do not remain FCC, α is not defined at all.

- The pairing is exact: same committee rows, same MD seeds, so member k differs between ensembles only by the constraint and the rejection step.
- **[TODO: if naive members fail to remain FCC, report that as a count rather than excluding them silently — “ α undefined for n of 5 members” is itself the result.]**

4.6 Grüneisen parameters

Mode Grüneisen parameters were computed by Eq. (21) from $\partial D/\partial V$ at $a_{\text{eq}}(1 \mp 0.01)$. Since phonopy was not available on this system, the branch-safe estimator is validated against a finite difference of sorted bands (Fig. 8): the two agree along the diagonal for almost every mode, with departures confined to a small number of points — exactly the band crossings at which the sorted-band derivative mis-assigns modes and Eq. (21) does not.

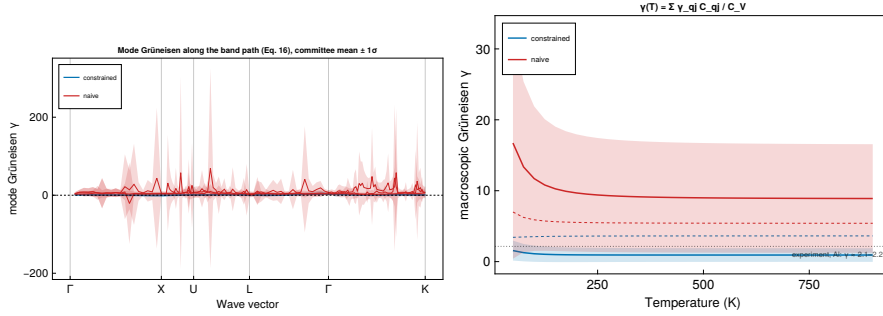


Figure 7: Left: mode Grüneisen parameter along the band path, ensemble mean $\pm 1\sigma$ for the constrained and naive committees (dashed: the respective point models). Right: macroscopic $\gamma(T) = \sum \gamma_{qj} C_{qj} / C_V$, with the experimental Al value $\gamma \approx 2.1-2.2$ marked.

Table 4: Macroscopic Grüneisen parameter at 300 K across the ensembles.

ensemble	$\gamma(300 \text{ K})$	range	$\gamma < 0$
naive POPS	9.24 ± 7.78	$[-2.97, 24.86]$	2/30
constrained + rejection	0.93 ± 0.97	$[-1.91, 2.30]$	4/30
experiment (Al)	2.1–2.2		

Points to make.

- The ensemble spread collapses by a factor of eight ($\sigma = 7.78 \rightarrow 0.97$). This is the same qualitative result as the phonon and elastic observables, on a third independent quantity.

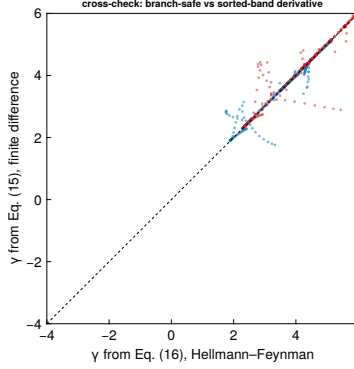


Figure 8: Validation of Eq. (21) (Hellmann–Feynman, branch-safe) against a finite difference of sorted bands. Agreement is exact except at band crossings, where the sorted-band derivative is known to fail.

- The constrained ensemble nonetheless *underestimates* γ (0.93 against 2.1–2.2), where the naive ensemble wildly overestimates it (9.24).
- **[TODO: decide how prominently to place the over-stiffening synthesis below — it is an honest limitation and arguably the most useful thing in the paper for a reader trying to apply the method.]**

4.7 A consistent over-stiffening, and what it implies

Three independent observables place the constrained ensemble on the same side of experiment:

Table 5: Constrained ensemble against experiment across three independent observables. All three indicate the same systematic error.

observable	measures	constrained	experiment	direction
α (Eq. 19)	anharmonicity	7.28×10^{-6}	$23\text{--}26 \times 10^{-6}$	too low
γ (Eq. 21)	anharmonicity	0.93	2.1–2.2	too low
γ_{us}	barrier height	360 mJ/m ²	160–180 mJ/m ²	too high

Points to make.

- These are not three separate shortcomings but one finding measured three ways: the constrained model is systematically too stiff. Two anharmonicity measures come out low and a barrier height comes out high, which is the signature of a potential-energy surface with too much curvature.
- The mechanism is visible in the constraint set. Eqs. (10), (11) and (16) all impose *signs* — positive curvature, positive elastic moduli, positive ω^2

— and never *magnitudes*. Nothing prevents the QP from satisfying every one of them by stiffening.

- This motivates constraints that pin a magnitude. Sec. 4.8 does exactly that for the elastic constants.
- **[TODO: state the scope honestly: the over-stiffening is measured on the 91-parameter A1.12.4.6A.2. model. Sec. 4.8 finds $\gamma_{\text{us}} \approx 175$ – 182 mJ/m² for the 5476-parameter A1.20.4.6A.4, i.e. within experiment, which suggests much of the effect is model capacity rather than the constraint scheme. Testing that properly means running the phonon-constrained pipeline on the larger model.]**

4.8 Pinning the elastic constants exactly

The constraints above impose elastic *stability* without constraining elastic *values*: across the constrained POPS ensemble, C_{11} spans 48–306 GPa while every member satisfies Eq. (11). Since the elastic operator is linear in c , pinning the constants exactly requires only that the Born rows be imposed as equalities, $l = u = C_{ij}^{\text{target}}$.

We test this on A1.20.4.6A.4 (5476 basis functions), comparing the regularised least-squares fit against one in which C_{11} , C_{12} and C_{44} are pinned to literature values while $b'(a_{\text{eq}}) \cdot c = 0$ is retained, so that both models sit at their own relaxed equilibrium.

Table 6: A1.20.4.6A.4: regularised least squares against exactly-pinned elastic constants. The equality constraints are attained to three decimal places, and a_{eq} is identical in both models, so the comparison is controlled. γ_{us} , G and ν are *not* constrained.

	nominal (RLS)	pinned C_{ij}	reference
a_{eq} (Å)	4.049236	4.049236	4.05
C_{11} (GPa)	116.67	116.300	target 116.3
C_{12} (GPa)	52.06	64.800	target 64.8
C_{44} (GPa)	36.70	30.900	target 30.9
G (GPa)	34.94	28.84	≈ 26
ν	0.295	0.3425	≈ 0.35
γ_{us} (mJ/m ²)	181.8	174.3	160–180
τ_P (GPa)	5.85×10^{-5}	1.56×10^{-4}	

Points to make.

- The unconstrained fit reproduces C_{11} to 0.3% but misses C_{12} by 20% (low) and C_{44} by 19% (high) — it gets the bulk-like combination right and the shear terms wrong. This is a concrete instance of the poorly-converged-DFT-stress argument from the introduction.

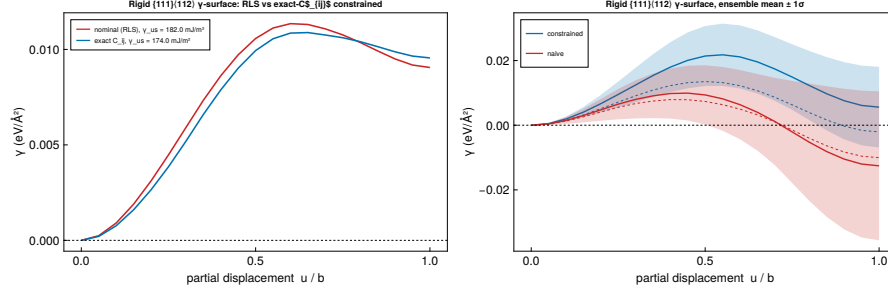


Figure 9: Left: rigid $\{111\}\langle 112 \rangle$ γ -surface for A1.20.4.6A.4, regularised least squares against exactly-pinned C_{ij} . Right: the same curve across the A1.12.4.6A.2. POPS ensembles, mean $\pm 1\sigma$.

- Pinning moves G and ν toward experiment, as it must. More interestingly it also improves γ_{us} ($181.8 \rightarrow 174.3$ mJ/m², further into the experimental band) — a quantity the constraint never touched. Elastic and stacking-fault physics are therefore coupled in the parameter space, and pinning one improves the other.
- τ_P shifts by a factor of 2.7. Since $\tau_P = f(\gamma_{us}, C_{ij})$ and the C_{ij} are now exact, the residual uncertainty in τ_P is attributable to γ_{us} alone.
- [TODO: the absolute τ_P values are small; the PN closed form used here is internally consistent but not literature-grade (see the module docstring). Present the *ratio*, not the absolute stress, or upgrade the PN treatment.]
- [TODO: natural next step for the ensemble: pin C_{ij} across a POPS committee and show the τ_P spread collapse. γ_{us} is itself a linear row (gsf_basis_delta.csv), so it could be pinned too.]

4.9 Uncertainty quantification: parity, calibration and coverage

On the training set the constrained-plus-rejection ensemble achieves 75.4% coverage against 70.7% for the naive hypercube *at essentially identical interval width* (median width ratio 0.987): the improvement is in placement, not in breadth.

Points to make.

- Always report coverage *with* interval width; coverage alone cannot distinguish a well-calibrated ensemble from a merely wider one.
- Construction C is built from only 30 observations and is therefore rank-limited — Eq. (9) spans at most 29 of 91 directions — and its training

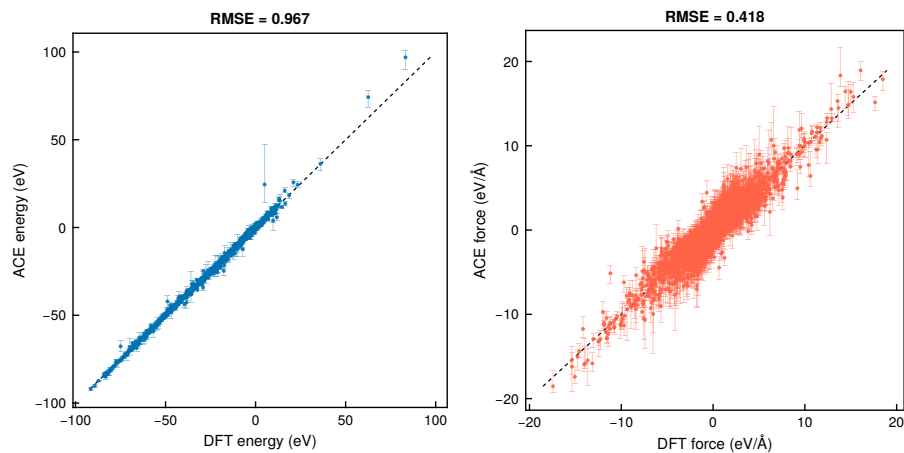


Figure 10: Test-set parity with ensemble min/max error bars. [TODO: state the stride and number of configurations used.]

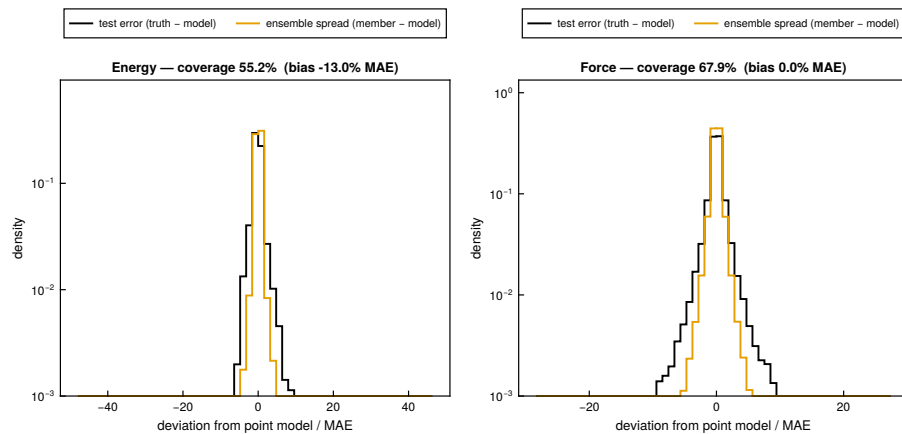


Figure 11: Calibration: the distribution of test error against the ensemble's own predictive spread, both normalised by the MAE. A well-calibrated ensemble has the two of comparable width.

Table 7: Test-set parity and calibration for the three ensemble constructions (1104 configurations, stride 20). Coverage is the fraction of test points inside the ensemble min/max envelope; width is the mean envelope width. Coverage per unit width is the meaningful comparison, since coverage alone rewards breadth.

construction	RMSE_E (eV)	cov_E (%)	width_E (eV)	$\text{cov}_E/\text{width}$ (%/eV)	cov_F (%)	width_F (eV/Å)
A: EOS-QP + phonon rejection	0.994	90.7	2.366	38.3	97.1	1.616
B: naive POPS, no constraints	0.989	93.6	2.447	38.2	99.2	1.874
C: full per-observation repair	0.994	25.9	0.379	68.4	29.9	0.220

coverage collapses to 29.8% with intervals $8\times$ narrower. This is a structural argument for constraining the whole cloud rather than a hand-picked subset, and is independent of any physics.

- Construction A matches B at *narrower* intervals: coverage per unit width is 38.3 against 38.2 %/eV on energies, and on forces A is better (97.1% at width 1.616 against 99.2% at 1.874, i.e. 14% narrower for 2 points less coverage). The constrained ensemble therefore gives naive-quality UQ *and* phonon-stable members — the central claim of the paper.
- A and C share an RMSE of 0.994 eV because RMSE is a property of the *point model* and both are centred on the constrained mean; B’s marginally lower 0.989 is the regularised least-squares model fitting slightly better. Worth stating explicitly — a referee will otherwise read the equality as a mistake.
- C’s high coverage-per-width (68.4) is an artefact of the rank deficiency, not a virtue: its envelope is so narrow that the ratio flatters it while absolute coverage collapses to 26%. This is the case where the per-width normalisation must *not* be read on its own.

4.10 Elastic stability across the ensembles

Evaluating the elastic constants member by member exposes a failure of naive POPS that is categorical rather than quantitative.

Table 8: Elastic behaviour across the **A1.12.4.6A.2.** ensembles. A member with $G \leq 0$ or $\nu \notin (-1, \frac{1}{2})$ is elastically unstable, and the Peierls–Nabarro prefactor $K = G/(1 - \nu)$ is then undefined — not merely inaccurate.

ensemble	C_{44} range (GPa)	G range (GPa)	elastically unstable
naive POPS	−54.5 – 66.2	−19.7 – 70.1	12/30
constrained + rejection	25.6 – 128.4	23.9 – 117.8	0/30

Points to make.

- Twelve of thirty naive members have a *negative shear modulus*. For those members τ_P is undefined, not large — the computation returns $-\infty$. The constrained ensemble has none, and every Poisson ratio lies in $(0.12, 0.30)$.
- This is a stronger statement than the phonon result, where at least a number came out for every member, and it lands on a dislocation-relevant quantity.
- The rigid γ -surface is itself a linear functional of c (Sec. 3.5), so γ_{us} for an entire ensemble costs one basis evaluation per displacement. Verified against direct energy evaluation to 1.7×10^{-14} eV/Å².
- [TODO: note that the constrained ensemble is elastically *stable* but not elastically *accurate* — C_{11} still spans 48–306 GPa. That is what Sec. 4.8 addresses.]

4.11 Computational cost

Points to make.

- Full per-observation cutting-plane repair costs ≈ 8.6 s per observation with ~ 800 accumulated constraint rows — ~ 176 core-hours for the top-50% leverage half of this dataset.
- The cheap variant replaces this with one small QP per observation and no cutting-plane loop, moving the physics into the sampler.
- Its dominant cost is instead the rejection acceptance rate; report the funnel, and note that a single volume ($s = 1.10$) accounted for 97.8% of rejections — which suggests adding only that volume’s rows to the QP as a middle path.
- [TODO: tabulate wall-clock: ensemble construction, per-member phonon check, *NPT* validation.]
- Frame the whole comparison against the alternative — generating more DFT — which is the motivating argument of the paper.

4.12 Secondary application: tungsten

[TODO: This subsection is yours — I have not worked on the W runs. Suggested structure below.]

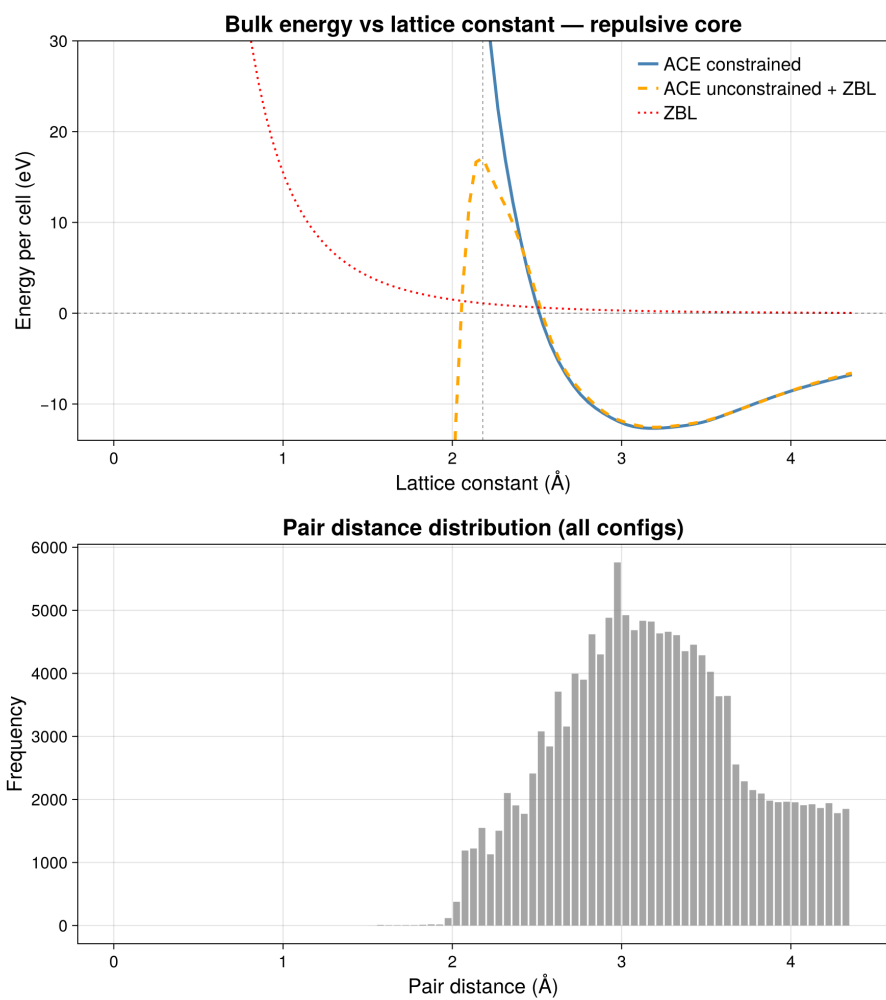


Figure 12: Bulk Tungsten cell energies for different lattice constants, with histogram detailing the distribution of pair distances in the training data set.

Points to make.

- Non-repulsive core behaviour outside the training distribution — tie directly to the pair-distance histogram in Fig. 12.
- Constraints formulated from experimental data rather than DFT, which addresses the poorly-converged-stress argument from the introduction.
- Access to high-pressure conditions the unconstrained model cannot represent.
- **[TODO: Pareto fronts (energy_pareto_front.png, force_pareto_front.png) — what is being traded against what, and where does the constrained model sit?]**